Algebra 1
Unit 6
Lesson 6 Practice

Name____________________
Date____________________
Period____________________

Jacqueline is buying grapes and apples from the Walmart produce section. Grapes are $\$3$ per pound and apples are $\$4$ per pound. Jacqueline's mother gave her $\$20$ to spend. To determine the combinations of grapes and apples that she can afford to purchase, Jacqueline wrote the inequality $3x + 4y \leq 20$.

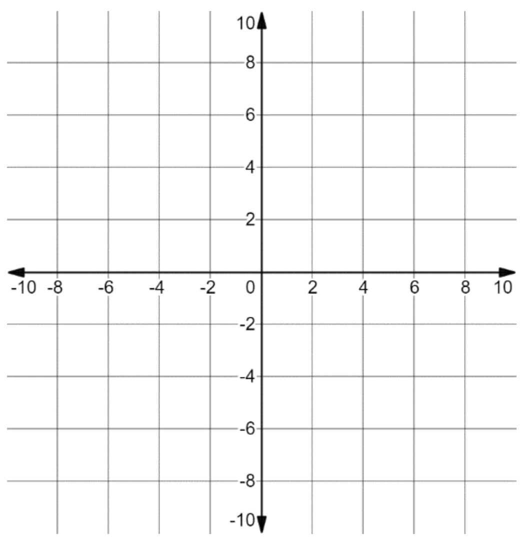
1. Define the variables Jacqueline used based on the context.

   Let x represent ____.

   Let y represent ____.

2. Explain why the inequality sign $\leq$ makes sense in this context.

3. Graph the inequality $3x + 4y \leq 20$.



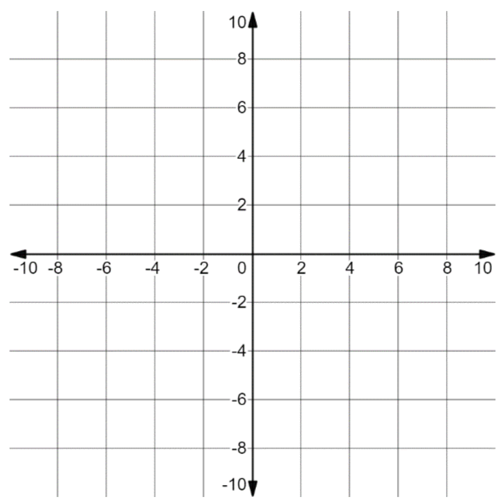
4. The ordered pair $(3, -4)$ is a solution to the inequality $3x + 4y \leq 20$. Explain why this is not a viable solution in the context of buying grapes and apples.

5. The ordered pair $(4, 2)$ is a point on the boundary line. Explain what the ordered pair represents in the situation.

Miguel invested some of his savings in the stock market. He bought 45 shares of a video production company and 60 shares of a clothing company. He hopes to make a combined profit of at least $\$360$ when he sells his shares. He wrote the inequality $45x + 60y \geq 360$ to represent the situation, where x is the change in value of one share of the video production company and y is the change in value of one share of the clothing company.

6. Explain what it would mean if x or y were a negative number.

7. Graph the inequality $45x + 60y \geq 360$.



8. Explain what the point $(-2.50, 9.75)$ represents in Miguel's situation.

9. Explain why it makes sense in Miguel's context that there are no solutions for which both x and y are both negative.

10. Stock prices in the United States are rounded to the nearest hundredths place. Complete the statement:

    In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be rounded to ____ decimal places.